

# Training Leaves Traces: Weight-Level Model Provenance Without Watermarks

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## Abstract

Model provenance depends on external evidence—hashes, metadata, logs, or watermarks—which can fail when checkpoints are adapted or records are incomplete. We ask whether *weight-level lineage* can be verified from weights alone: given a reference checkpoint and a suspect checkpoint of compatible architecture, did the suspect inherit the reference’s weights under post-training transformations? We find that modern residual architectures expose such a signal in their architecture-aware branch products. In a two-layer residual branch with input projection  $W_{\text{in}}$  and output projection  $W_{\text{out}}$ , training couples paired matrices so that  $W_{\text{out}}W_{\text{in}}$  exhibits a diagonal-dominance fingerprint absent from mismatched or random pairings; for deeper or non-MLP branches we measure the trace on the architecture-appropriate product (e.g.  $W_3W_2W_1$  in Bottleneck ResNets,  $W_OW_V$  and  $W_QW_K^T$  in attention, joint SwiGLU products), formalize it with  $s(M) = |\text{tr}(M)|/\|M\|_F$ , derive a separation margin for the residual-MLP regime, and use a two-level pipeline: diagonal dominance detects trained branch structure for block pairing, and centered residual signatures detect checkpoint lineage across pairs of models.

Empirically, across 12 model architectures (GPT-2 124M–1.5B, BERT-base, Mistral-7B, LLaMA-2-7B base and RLHF chat, Qwen2.5-7B, DeepSeek-R1-Distill-Llama-8B, Whisper tiny/base/small, ResNet-18/50/101/152, ViT-B/16, and ConvNeXt-T) spanning 5 architectural families, 5 post-training weight modifications, and over 280 reference–suspect pairs, the fingerprint recovers 91–100% of residual-block correspondences and the centered residual-signature lineage score separates descendants from non-descendants at AUROC = 1.000 with TPR = 100% at the 1% false-positive operating point on both synthetic depth-24 MLP benchmarks and real ResNet-18/CIFAR-10 checkpoints. The fingerprint is absent at initialization, appears within the first five epochs of training, follows the residual-MLP margin formula’s  $\sqrt{d}$  envelope on GPT-2 MLPs, and persists under fine-tuning, quantization, pruning, weight noise, RLHF, and reasoning-trace distillation. In direct gradient-based suppression experiments, reducing the lineage score below the descendant threshold incurs eval-loss increases of 12–100%. Boundary cases clarify the signal: independently trained same-architecture models do not falsely match ( $\mathcal{L} \leq 0.20$ ), while knowledge-distilled students remain near the null distribution ( $\mathcal{L} \approx 0.086$ ) despite functional imitation—correctly classified as non-descendants because they share no weights.

## 1 Introduction

The open-weight ecosystem has grown explosively: Hugging Face now hosts nearly one million model repositories, with derivatives proliferating faster than originals (Hugging Face 2024). When Meta’s LLaMA weights leaked in March 2023, the community produced instruction-tuned variants—Alpaca, Vicuna, Koala—within weeks, each costing under \$300 to create (Touvron et al. 2023). This rapid, low-cost adaptation means any widely-distributed checkpoint spawns hundreds of fine-tuned, merged, quantized, and renamed descendants whose lineage quickly becomes untraceable.

Current approaches to model provenance rely on externally maintained evidence. Model cards document training datasets and licensing terms (Mitchell et al. 2019); experiment tracking platforms such as MLflow and Weights & Biases record artifacts and lineage graphs; cryptographic hashes provide integrity checks for unmodified files. For cases where theft is anticipated, watermarking methods embed ownership signals into network parameters (Uchida et al. 2017) or decision boundaries (Adi et al. 2018) that can later be extracted to verify provenance.

Yet each approach has fundamental limitations. Cryptographic hashes become invalid after any weight modification—including routine quantization or fine-tuning. Metadata and provenance logs can be falsified, lost, or simply never recorded. Watermarks require deliberate insertion before deployment and remain vulnerable to removal: a recent systematization found that no surveyed DNN watermarking scheme proved robust against fine-tuning, pruning, or model extraction attacks (Lukas et al. 2022). These limitations leave a critical gap: no established method provides robust white-box weight-level lineage verification for post-hoc, modified checkpoints when no watermark was inserted before deployment.

Prior intrinsic fingerprinting methods have asked whether weight statistics—means, variances, or histogram features—can identify models (Zheng et al. 2022; Zhao et al. 2020), but these aggregate properties lack verification precision and are easily disrupted. Training dynamics research has established that gradient descent enforces near-orthogonal Jacobians in residual networks (Pennington, Schoenholz, and Ganguli 2017; Saxe, McClelland, and Ganguli 2014), yet this work focused on trainability, not on what structural traces the process leaves behind. We ask a different question:

does training itself imprint a recoverable signature—not in aggregate statistics, but in the relational structure between paired weight matrices?

The answer is yes. In residual architectures, trained architecture-aware branch products develop characteristic trace/Frobenius concentration: diagonal mass that is absent from mismatched or random pairings. This phenomenon emerges because residual blocks must maintain near-identity behavior for stable gradient flow—a requirement formalized as *dynamical isometry*, where the input-output Jacobian’s singular values concentrate near unity throughout training (Pennington, Schoenholz, and Ganguli 2017; Saxe, McClelland, and Ganguli 2014). In two-factor residual MLPs ( $x + W_{\text{out}}\phi(W_{\text{in}}x)$ ), this constraint pulls the branch product toward a negative-identity-like component,  $W_{\text{out}}W_{\text{in}} \approx -\varepsilon I$  (He et al. 2016), and admits a closed-form margin analysis. In multi-factor and attention-style branches (e.g.,  $W_3W_2W_1$ ,  $W_OW_V$ ,  $W_QW_K^T$ , SwiGLU joint products), the same training pressure produces trace concentration in the architecture-appropriate product without necessarily taking the strict negative-identity form. We find this fingerprint is absent at initialization, emerges within the first five epochs of training, and yields a diagonal-dominance score whose separation margin scales as  $\Theta(\sqrt{d})$  in the residual-MLP regime—an envelope that GPT-2 MLPs follow empirically.

This trace enables *passive white-box weight-level lineage verification*: given a reference checkpoint and a suspect checkpoint of compatible architecture, decide whether the suspect inherited the reference’s weights under post-training transformations, without watermarks, metadata, or cooperation from the original trainer. Unlike embedded watermarks that must be inserted before or during training, the diagonal-dominance fingerprint emerges from optimization itself and can be read retroactively from compatible residual branch products in already-deployed checkpoints—including models released long before provenance verification was anticipated. The fingerprint maintains 100% block-pair accuracy across 21 attack configurations (fine-tuning, quantization, pruning, weight noise up to  $\sim 20\%$ ), degrading only when perturbations also destroy model utility. Boundary cases clarify the signal’s scope: independently trained models do not falsely match, while distillation removes the trace—correctly, since the method detects weight-level lineage rather than functional imitation.

**Two tasks, one mechanism.** The verification problem decomposes into two distinct sub-problems that we treat separately throughout. *Block correspondence* asks, within a single model, which  $W_{\text{in}}$  pairs with which  $W_{\text{out}}$ ; the diagonal-dominance score  $s(M) = |\text{tr}(M)|/\|M\|_F$  on the architecture-aware product is sufficient for this, and we report pair-accuracy numbers throughout. *Checkpoint lineage* asks, across two checkpoints of compatible architecture, whether one descends from the other; this requires subtracting the generic identity-like component before comparison and is reported as AUROC. Diagonal dominance detects trained branch structure; centered residual signatures detect checkpoint lineage. Both tasks share the same underlying mechanism, but they have different inputs, outputs, and failure modes, and they should not be conflated.

We make the following contributions:

- **We formalize the diagonal-dominance signal and derive a separation margin that grows with width.** Building on Park (2026)’s observation that training induces negative diagonal structure in residual-branch products, we define the score  $s(i, j) = |\text{tr}(M)|/\|M\|_F$  and prove that, in the residual-MLP regime captured by Proposition 1, it achieves  $\Theta(\sqrt{d})$  for correct pairs versus  $\Theta(1/\sqrt{d})$  for mismatched pairs; GPT-2 MLPs follow this  $\sqrt{d}$  envelope empirically (Section 3).
- **We develop a passive provenance verification protocol.** Cross-block score matrices combined with Hungarian matching recover residual-branch correspondences and yield checkpoint-level ancestry evidence from weights alone, for suspect checkpoints of compatible architecture—without watermarks, metadata, or cooperation from the original trainer (Section 4).
- **The fingerprint generalizes across architectures and survives post-training modification.** We validate across nine architecture families (GPT-2 to 1.5B, BERT, LLaMA, Mistral, Qwen, Gemma, ViT, Whisper, ResNet), demonstrate robustness under 21 attack configurations, and confirm via ablation that the signal requires both training and residual connections (Sections 5–6).

## 2 Background and Problem Setting

### 2.1 Passive Model Provenance

We define *passive model provenance* as follows: given a reference checkpoint  $R$  and a suspect checkpoint  $S$  with missing or untrusted metadata, determine whether  $S$  shares weight-level lineage with  $R$  after allowed post-training transformations. This is a *white-box* verification task—the verifier has access to both sets of weights. The setting matches regulatory audits, model hub compliance checks, and ownership disputes where both parties can produce their checkpoints.

The method detects *weight-level lineage*, not behavioral equivalence. Two models share weight-level lineage if the suspect’s parameters can be derived from the reference through transformations such as fine-tuning, pruning, or quantization—without retraining from scratch. This is distinct from functional equivalence, where models produce similar outputs regardless of whether their weights share any relationship. Model extraction attacks (Tramèr et al. 2016) create functional copies with entirely fresh weights; such models are behaviorally similar but share no weight-level identity.

Distilled models and independently trained functional copies are explicitly outside the positive guarantee. A student trained via knowledge distillation (Hinton, Vinyals, and Dean 2015) may replicate the teacher’s behavior with high fidelity, but it evolves its weights through a separate optimization process that produces distinct structural fingerprints. This boundary is intentional: proving that weights were copied or derived requires distinguishing “same functionality” from “same parameters (up to transformation).”

## 2.2 Limits of Existing Methods

Model provenance verification has relied on three families of approaches, each with fundamental limitations.

**Cryptographic hashes** fail immediately upon any weight modification. The avalanche effect ensures that fine-tuning, pruning, or even numerical precision changes produce entirely different digests, providing no signal of derivation.

**Metadata and logging systems**—model cards (Mitchell et al. 2019), experiment trackers like MLflow and Weights & Biases—require trusted provenance records that may be incomplete, falsified, or absent when models are redistributed outside controlled environments.

**Watermarking methods** embed ownership signals into weights (Uchida et al. 2017; Rouhani, Chen, and Koushanfar 2019) or train models to produce specific outputs on trigger inputs (Adi et al. 2018; Zhang et al. 2018). Both require forethought: the watermark must be inserted before training concludes, precluding retroactive verification of already-deployed models. A systematic evaluation found that no surveyed watermarking scheme is robust in practice against fine-tuning, pruning, and model extraction attacks (Lukas et al. 2022).

**Output watermarking** systems such as SynthID (Dathathri et al. 2024) identify AI-generated content by embedding statistical signatures during generation, but address a fundamentally different question: they detect whether content was AI-generated, not whether one model’s weights derive from another’s.

**Decision-boundary fingerprinting** (Cao, Jia, and Gong 2021; Le Merrer, Perez, and Trédan 2020) identifies functional similarity via adversarial examples near classification boundaries, but cannot distinguish weight lineage: a distilled student may share decision surfaces while having entirely independent weights.

The key differentiator of our approach: prior watermarking work asks how to *insert or protect* an ownership signal. We ask whether training already *leaves* a signal that can be read retroactively.

## 2.3 Residual Branch Products

Modern deep networks use *residual connections*—shortcuts that allow information to bypass layers—to enable stable training of very deep architectures (He et al. 2016). A residual block computes  $x' = x + F(x)$ , where  $x$  is the input and  $F$  is the residual branch function. For these blocks to maintain stable gradient flow, the Jacobian  $J = I + J_F$  must remain close to orthogonal throughout training—a condition formalized as *dynamical isometry* (Pennington, Schoenholz, and Ganguli 2017; Saxe, McClelland, and Ganguli 2014).

For a residual branch with  $K$  linear transformations separated by nonlinearities, we define the *residual branch product*:

$$M = W_K W_{K-1} \cdots W_1 \in \mathbb{R}^{d \times d} \quad (1)$$

This product captures how the branch transforms the input space. In the two-factor residual-MLP regime, the dynamical isometry constraint admits a clean characterization:  $M$  approximates a negative scaled identity,  $M \approx -\varepsilon I$  for small  $\varepsilon > 0$  (Tarnowski et al. 2019; De and Smith

| Architecture     | Residual Form                   | Product $M$    |
|------------------|---------------------------------|----------------|
| BasicBlock / MLP | $x + W_2 \phi(W_1 x)$           | $W_2 W_1$      |
| Transformer MLP  | $x + W_2 \phi(W_1 x)$           | $W_2 W_1$      |
| Attention V/O    | $x + W_O \text{Attn}(W_V x)$    | $W_O W_V$      |
| Attention Q/K    | $\text{softmax}(QK^\top)$       | $W_Q W_K^\top$ |
| Bottleneck       | $x + W_3 \phi(W_2 \phi(W_1 x))$ | $W_3 W_2 W_1$  |
| SwiGLU MLP       | gated linear unit               | $W_d W_u$      |

Table 1: Architecture-aware factorization for residual branch products.

2020). We use this as the working model for our margin analysis; the broader empirical observation that trained branch products exhibit trace/Frobenius concentration extends to multi-factor and attention-style products even when the strict negative-identity geometry does not hold.

Table 1 shows the architecture-aware factorization for common residual blocks. Using the correct factorization is critical: the naive endpoint product  $W_3 W_1$  for Bottleneck ResNets (skipping the middle convolution) yields chance-level pairing accuracy, while the full triple product  $W_3 W_2 W_1$  recovers 91–100% of correct pairs.

## 3 Training-Induced Diagonal Dominance

### 3.1 Empirical Observation

Training a deep residual network imprints a characteristic structure on its weights. We first demonstrate this phenomenon empirically before formalizing it.

**At initialization, no diagonal structure appears.** We compute the score matrix  $s(i, j)$  for all pairs of input and output projections in a 48-block residual network. At epoch 0, the matrix shows no discernible pattern: Hungarian matching recovers only 6.25% of correct pairs (chance is  $1/48 \approx 2.1\%$ ), and pair separation—the gap between the worst correct-pair score and the best incorrect-pair score—is  $-0.42$  (negative indicates no separation). Across seven initialization schemes—Kaiming uniform/normal, Xavier uniform/normal, Gaussian, uniform, and orthogonal—pair accuracy remains at chance level ( $\leq 2.1\%$ ). Critically, orthogonal initialization provides dynamical isometry at initialization by construction (Saxe, McClelland, and Ganguli 2014), yet shows *zero* correct pairs before training, ruling out the hypothesis that near-orthogonal Jacobians alone create the fingerprint.

**During training, diagonal structure emerges rapidly.** By epoch 5, the diagonal of  $s(i, j)$  is fully separated and Hungarian matching recovers 100% of correct pairs. The mean correct-pair score rises monotonically from 0.19 at epoch 0 to 2.07 at epoch 300, while the mean incorrect-pair score remains flat at  $\approx 0.16$ —a  $21\times$  ratio at convergence. The transformation occurs within the first few epochs, well before training loss plateaus.

**The effect is not explained by matrix shape or architecture alone.** A null model with randomly initialized weights achieves only 6.25% pair accuracy with negative separation, ruling out shape compatibility as an explanation.

To confirm the fingerprint requires residual training specifically, we train a PlainNet—architecturally identical except that skip connections are removed. While both networks converge to similar evaluation loss (1.35 vs 1.56), ResNet achieves 100% pair accuracy (AUC 0.98) with 100% of correctly paired products exhibiting negative trace, while PlainNet achieves only 3%—at the chance baseline of  $1/24 \approx 4.2\%$ —with AUC 0.51 and no discernible diagonal structure. The skip connection creates a functional constraint—the block must act as a near-identity perturbation to preserve gradient flow—that imprints the characteristic trace structure.

### 3.2 Diagonal-Dominance Fingerprint

For a candidate residual branch product  $M_{ij} = W_{\text{out}}^{(j)} W_{\text{in}}^{(i)}$ , we define the *diagonal-dominance score*:

$$s(i, j) = \frac{|\text{tr}(M_{ij})|}{\|M_{ij}\|_F} \quad (2)$$

The numerator  $|\text{tr}(M)|$  captures *diagonal mass*—how much of the matrix energy concentrates on the diagonal. The denominator  $\|M\|_F$  normalizes by total matrix energy (Frobenius norm), isolating structural fingerprint from scale.

This ratio has a natural geometric interpretation. A matrix with energy uniformly spread across entries achieves  $s = \Theta(1/\sqrt{d})$ , while a matrix proportional to the identity achieves  $s = \sqrt{d}$ . The key insight is that correctly paired blocks—where all  $K$  matrices come from the same residual branch—score at  $\Theta(\sqrt{d})$ , while incorrect pairings remain at the random baseline  $\Theta(1/\sqrt{d})$ . This yields a signal-to-noise gap that grows linearly with hidden dimension  $d$ , making correct pairs increasingly separable in wider networks.

Dynamical isometry (Pennington, Schoenholz, and Ganguli 2017) predicts that a well-trained residual block has Jacobian  $J = I + W_{\text{out}} W_{\text{in}}$  close to orthogonal. In the two-factor residual-MLP regime, the cleanest realization of this constraint is for the branch product to approximate a negative scaled identity, and we adopt this as our working model for the margin analysis. We formalize it with the decomposition:

$$M = W_{\text{out}}^{(i)} W_{\text{in}}^{(i)} = -\varepsilon I + E \quad (3)$$

where  $\varepsilon = |\text{tr}(M)|/d > 0$  captures diagonal strength and  $E$  is the residual with  $\text{tr}(E) = 0$ . The matrix  $E$  contains all off-diagonal structure plus the zero-trace portion of the diagonal. Dynamical isometry implies  $\|E\|_F \ll \varepsilon\sqrt{d}$  for well-trained blocks; the following proposition quantifies the margin exactly. The decomposition is exact for any matrix, but its predictive power depends on the regime: in two-factor residual MLPs the negative-identity component dominates, while in multi-factor and attention-style products (Section 5) the trained product retains the same trace-concentration signal even when  $E$  is too large for the negative-identity geometry to hold cleanly.

Crucially, diagonal dominance captures *relational* structure between paired matrices—how they jointly encode the near-identity mapping—rather than aggregate properties of individual matrices. This relational fingerprint is preserved

under fine-tuning and noise because it reflects the functional constraint that the residual block must approximate.

### 3.3 Margin Analysis and Null Model

**Proposition 1** (Margin Formula). *Let  $M = -\varepsilon I + E$  with  $\varepsilon = |\text{tr}(M)|/d$  and  $\text{tr}(E) = 0$ . Then the diagonal-dominance score satisfies*

$$s(i, i) = \frac{\sqrt{d}}{\sqrt{1 + \|E\|_F^2/(\varepsilon^2 d)}} \leq \sqrt{d}, \quad (4)$$

with equality if and only if  $E = 0$ , i.e.,  $M = -\varepsilon I$ .

*Proof.* By construction,  $|\text{tr}(M)| = \varepsilon d$ . For the Frobenius norm, expand  $\|M\|_F^2 = \|- \varepsilon I + E\|_F^2 = \varepsilon^2 d - 2\varepsilon \text{tr}(E) + \|E\|_F^2 = \varepsilon^2 d + \|E\|_F^2$ , where the last step uses  $\text{tr}(E) = 0$ . Substituting into (2):  $s(i, i) = \varepsilon d / \sqrt{\varepsilon^2 d + \|E\|_F^2} = \sqrt{d} / \sqrt{1 + \|E\|_F^2/(\varepsilon^2 d)}$ . The bound follows immediately, with equality iff  $\|E\|_F = 0$ .  $\square$

The margin formula reveals the geometry of diagonal dominance. The numerator  $\sqrt{d}$  is the score achievable by a perfect negative-scalar identity; the denominator penalizes deviation from this ideal via the dimensionless ratio  $\|E\|_F^2/(\varepsilon^2 d)$ . For typical trained blocks,  $\|E\|_F/(\varepsilon\sqrt{d}) \in [1.9, 3.8]$ , yielding  $s(i, i) \in [1.76, 3.23]$ —well above the baseline but below the theoretical maximum of  $\sqrt{d}$ .

**Corollary 1** (Signal-to-Baseline Ratio). *Suppose  $(W_{\text{in}}^{(i)}, W_{\text{out}}^{(j)})$  with  $i \neq j$  are independent random matrices with i.i.d. zero-mean entries. Then  $\mathbb{E}[s(i, j)^2] = 1/d$ , so  $\mathbb{E}[s(i, j)] \approx 1/\sqrt{d}$ . The signal-to-baseline ratio scales as  $d$ .*

Correct pairs achieve scores of order  $\sqrt{d}$ , while incorrect pairs score at the random baseline  $1/\sqrt{d}$ . The gap grows with dimension, providing increasingly robust separation for wider networks. On GPT-2 models, mean  $s(i, i)$  rises from 4.18 ( $d = 768$ , 124M parameters) to 7.77 ( $d = 1600$ , 1.5B parameters), consistent with  $\sqrt{d}$  scaling.

**Corollary 2** (Null Model). *For randomly initialized weights with no training, all pairs satisfy  $\mathbb{E}[s(i, j)] = 1/\sqrt{d} + o(1)$  uniformly in  $(i, j)$ . Hungarian matching recovers correct pairs at the chance rate  $1/L$ .*

The null model rules out three alternative explanations: (a) diagonal dominance is not an artifact of compatible matrix shapes; (b) the residual connection itself does not create the signal; (c) any nontrivial loss level does not suffice. The diagonal-dominance fingerprint is a *learned* property induced by gradient descent. The theory explains why the observed structure is identifiable once induced; controlled experiments—including the initialization ablation showing orthogonal init at 0% pairing and the PlainNet control at 3%—confirm that training induces it

## 4 From Fingerprints to Provenance Verification

This section develops the computational machinery for transforming the diagonal-dominance fingerprint into a

practical provenance verification system. We present: (1) an algorithm for recovering residual-block correspondences between checkpoints, (2) a model-level lineage score with statistical calibration, and (3) an end-to-end verification protocol suitable for deployment.

#### 4.1 Pairing Algorithm

Given a reference model  $R$  with  $L$  residual blocks and a suspect model  $S$  with  $L$  blocks, we seek to determine which block in  $S$  corresponds to which block in  $R$ . Let  $\{(W_{\text{in}}^{(i)}, W_{\text{out}}^{(i)})\}_{i=1}^L$  denote the input and output projections of  $R$ 's residual branches, and  $\{(\tilde{W}_{\text{in}}^{(j)}, \tilde{W}_{\text{out}}^{(j)})\}_{j=1}^L$  denote those of  $S$ . If  $S$  descends from  $R$ —through fine-tuning, quantization, or other post-training modifications—then a permutation  $\pi : [L] \rightarrow [L]$  should exist such that block  $j$  in  $S$  corresponds to block  $\pi(j)$  in  $R$ .

**Score Matrix Construction.** We construct a score matrix  $\mathbf{S} \in \mathbb{R}^{L \times L}$  where each entry quantifies the diagonal-dominance evidence that block  $i$  in the reference corresponds to block  $j$  in the suspect:

$$s(i, j) = \frac{|\text{tr}(M_{ij})|}{\|M_{ij}\|_F}, \quad \text{where } M_{ij} = \tilde{W}_{\text{out}}^{(j)} W_{\text{in}}^{(i)}. \quad (5)$$

The cross-model product  $M_{ij}$  tests whether the output projection from suspect block  $j$  and the input projection from reference block  $i$  exhibit the trained coupling characteristic of a true residual branch. If  $S$  descends from  $R$  and block  $j$  corresponds to block  $i$ , then  $M_{ij}$  inherits the diagonal-dominant structure of the original branch product, yielding  $s(i, j) = \Theta(\sqrt{d})$ . For non-corresponding pairs, the matrices lack this coupling, and  $s(i, j)$  remains at the random baseline  $\Theta(1/\sqrt{d})$ .

**Optimal Assignment.** Given the score matrix  $\mathbf{S}$ , we seek a one-to-one assignment  $\pi : [L] \rightarrow [L]$  that maximizes total evidence for correct correspondences. This is the classical optimal assignment problem, solvable via the Hungarian algorithm (Kuhn 1955):

$$\pi^* = \arg \max_{\pi \in \mathcal{P}_L} \sum_{i=1}^L s(i, \pi(i)), \quad (6)$$

where  $\mathcal{P}_L$  denotes the set of all permutations on  $[L]$ . For derived models with preserved block ordering, the optimal assignment recovers the identity permutation:  $\pi^*(i) = i$  for all  $i \in [L]$ .

**Complexity.** Score matrix computation requires  $O(L^2 d^2 h)$  operations, where  $h$  is the intermediate dimension (typically  $4d$  for transformer MLPs). The Hungarian algorithm runs in  $O(L^3)$ . For GPT-2-XL ( $L = 48$ ,  $d = 1600$ ), verification completes in approximately 10 seconds on a single GPU.

**Empirical Validation.** On GPT-2 ( $L = 12$ ,  $d = 768$ ), the pairing algorithm achieves 100% pair accuracy with mean diagonal score  $\bar{s}(i, i) = 4.18$  and mean off-diagonal score  $\bar{s}(i, j) = 0.12$  for  $i \neq j$ —a  $35\times$  separation ratio. At random initialization, pair accuracy drops to 6.25% (chance level). The algorithm relates to permutation-based alignment methods for model merging (Ainsworth, Hayase, and Srinivasa

2023) and federated learning (Singh and Jaggi 2020), but differs in objective: we match blocks to establish lineage evidence rather than to enable weight interpolation.

#### 4.2 Model-Level Lineage Score

Block-level pairing establishes correspondences, but provenance decisions require a scalar statistic. We aggregate block-level similarity into a model-level lineage score with statistical calibration.

**Residual Signature.** For each branch  $\ell$ , we compute the residual signature that isolates checkpoint-specific structure by subtracting the generic diagonal-dominant component:

$$\alpha_\ell^* = \frac{\text{tr}(M_\ell)}{d}, \quad R_\ell = M_\ell - \alpha_\ell^* I, \quad \phi_\ell(M) = \frac{\text{vec}(R_\ell)}{\|\text{vec}(R_\ell)\|_2}, \quad (7)$$

where  $M_\ell = W_{\text{out}}^{(\ell)} W_{\text{in}}^{(\ell)}$  is the branch product and  $\alpha_\ell^*$  is the best Frobenius-fit identity scalar. The matrix  $R_\ell$  is exactly the branch-specific error term  $E_\ell$  from the dynamic-isometry decomposition  $M_\ell \approx -\varepsilon I + E_\ell$ ; subtracting  $\alpha^* I$  removes the generic component shared across all trained residual models and isolates the lineage signal.

**Gated Branch Score.** Branch similarity is reliable only when both branches are trained strongly enough to exhibit a valid diagonal-dominance fingerprint. We gate the cosine similarity by the diagonal-dominance score  $s(M) = |\text{tr}(M)|/\|M\|_F$ :

$$G(\phi_\ell^A, \phi_m^B) = \langle \phi_\ell^A, \phi_m^B \rangle \cdot \min\left(\frac{s(M_\ell^A)}{\tau_s}, \frac{s(M_m^B)}{\tau_s}, 1\right), \quad (8)$$

where the threshold  $\tau_s$  is the minimum diagonal-dominance score across reference branches (so every branch of a trained reference passes the gate at strength 1).

**Lineage Score.** The model-level lineage score between checkpoints  $A$  and  $B$  is the Hungarian-aligned mean of gated branch scores:

$$\mathcal{L}(A, B) = \frac{1}{L} \sum_{\ell=1}^L G(\phi_\ell^A, \phi_{\pi(\ell)}^B), \quad (9)$$

$$\hat{\pi} = \arg \max_{\pi \in \mathcal{S}_L} \sum_{\ell} G(\phi_\ell^A, \phi_{\pi(\ell)}^B).$$

This score satisfies: (1) self-similarity  $\mathcal{L}(A, A) = 1$ ; (2) symmetry  $\mathcal{L}(A, B) = \mathcal{L}(B, A)$ ; (3) boundedness  $\mathcal{L}(A, B) \in [-1, 1]$ ; and (4) orthogonality  $\mathbb{E}[\mathcal{L}(A, B)] \approx 0$  when  $A$  and  $B$  are independently trained.

**Statistical Calibration.** We calibrate against a null distribution of unrelated models. Let  $\mathcal{N}_A = \{\mathcal{L}(A, U_j) : U_j \notin \mathcal{D}(A)\}_{j=1}^n$  be scores between reference  $A$  and independently trained models  $U_j$ . The calibrated z-score is:

$$Z(A, B) = \frac{\mathcal{L}(A, B) - \mu_{\text{null}}}{\sigma_{\text{null}}}, \quad (10)$$

where  $\mu_{\text{null}}$  and  $\sigma_{\text{null}}$  are the mean and standard deviation of  $\mathcal{N}_A$ .

**Empirical Score Distributions.** On a controlled benchmark of depth-24 residual MLPs (3 reference checkpoints;

| Suspect Type                                       | $n$ | Mean $\mathcal{L}$ | Min   | Max   |
|----------------------------------------------------|-----|--------------------|-------|-------|
| Fine-tune (same task)                              | 15  | 0.996              | 0.990 | 1.000 |
| Quantization (16–256 lvl)                          | 15  | 0.995              | 0.980 | 1.000 |
| Gaussian noise ( $\sigma_{\text{rel}} \leq 15\%$ ) | 15  | 0.993              | 0.974 | 1.000 |
| Fine-tune (different task)                         | 15  | 0.944              | 0.840 | 0.998 |
| Magnitude pruning (10–85%)                         | 15  | 0.810              | 0.575 | 0.997 |
| Independent (same task)                            | 45  | 0.084              | 0.022 | 0.198 |
| Independent (diff. task)                           | 15  | 0.073              | 0.011 | 0.156 |
| Random init                                        | 15  | 0.070              | 0.018 | 0.139 |
| Distilled student                                  | 9   | 0.086              | 0.030 | 0.187 |

Table 2: Lineage score distributions by suspect category. Descendants cluster near  $\mathcal{L} = 1$ ; all non-descendant classes—including the strongest negative control, distilled students that are function-similar to the reference—remain near  $\mathcal{L} = 0.08$ . The separation between weakest descendant (pruning, 0.81 mean) and strongest non-descendant (distilled, 0.086 mean) is  $\approx 0.72$ .

75 descendants spanning fine-tuning, Gaussian noise, magnitude pruning, and quantization; 84 non-descendants spanning independent same-task training, independent different-task training, random initialization, and distilled students), Table 2 reports lineage scores by suspect category.

The separation is striking: descendants achieve  $\mathcal{L} \geq 0.81$  even under aggressive transformations, while non-descendants remain below  $\mathcal{L} < 0.20$ . The proposed residual-signature score yields perfect classification (AUROC = 1.000, AUPRC = 0.987, TPR@1%FPR = 100%, TPR@10%FPR = 100%). Critically, a baseline that uses only the diagonal-dominance score  $s(M)$ —which detects whether a model is trained but not whether it descends from a particular checkpoint—achieves AUROC 0.417 (chance level), confirming that the residual signature contributes the lineage information beyond generic dynamical-isometry structure.

**Lineage Chain and Branching Tree.** To test whether  $\mathcal{L}$  recovers multi-generational ancestry, we construct a branching tree: a root  $C_1$  trained from scratch, two siblings  $C_2, C_3$  fine-tuned from  $C_1$  on drifted targets, and two grandchildren  $C_4 = C_2 \rightarrow \text{fine-tune}$ ,  $C_5 = C_3 \rightarrow \text{fine-tune}$ . The lineage score recovers the correct ancestry for both leaves: for  $C_4$ , the ranking by  $\mathcal{L}(C_4, \cdot)$  is  $C_2$  (parent, 0.96)  $>$   $C_1$  (grandparent, 0.91)  $>$   $C_3$  (uncle, 0.87)  $>$   $C_5$  (cousin, 0.85)  $>$  5 independent same-task models (0.07–0.09); the analogous ranking holds for  $C_5$ . The score decays monotonically with genealogical distance—direct parent 0.95, grandparent 0.91, sibling 0.90, uncle 0.87, cousin 0.85, independent 0.08—while the family/strangers gap stays at  $\approx 10\times$ . This demonstrates that  $\mathcal{L}$  supports not just binary descendant-detection but compositional reasoning about checkpoint ancestry.

### 4.3 Verification Protocol

We formalize the end-to-end verification procedure. The protocol takes a reference checkpoint  $A$ , suspect checkpoint  $B$ , and pre-calibrated null distribution  $\mathcal{N}$ , outputting one of four verdicts.

| Transformation | Parameters               | Detection |
|----------------|--------------------------|-----------|
| Fine-tuning    | LR $10^{-5}$ – $10^{-3}$ | 100%      |
| Quantization   | 16-bit, 8-bit            | 100%      |
| Quantization   | 6-bit                    | 94%       |
| Gaussian noise | $\sigma \leq 0.3$        | 100%      |
| Gaussian noise | $\sigma = 0.5$           | 96%       |
| Pruning        | up to 30%                | 100%      |
| LoRA merge     | rank 8–64                | 100%      |

Table 3: Verification robustness. Detection rate is fraction of true descendants correctly identified at  $\tau = 1.645$ .

#### Protocol Steps:

1. **Compatibility Check:** Verify  $\text{arch}(A) = \text{arch}(B)$ . If not, return INCOMPATIBLE.
2. **Extract:** Compute architecture-aware branch products  $\{M_\ell^A\}, \{M_\ell^B\}$  using Table 1.
3. **Fingerprint:** Compute centered diagonal fingerprints  $\{\psi_\ell^A\}, \{\psi_\ell^B\}$ .
4. **Align:** If block correspondence is unknown, apply Hungarian matching (Section 4.1).
5. **Aggregate:** Compute lineage score  $\mathcal{L}(A, B)$  via Eq. (9).
6. **Calibrate:** Compute z-score  $Z(A, B)$  via Eq. (10).
7. **Decide:** Return verdict based on thresholds  $\tau_{\text{upper}}, \tau_{\text{lower}}$ :

$$\text{Verdict} = \begin{cases} \text{DESCENDANT} & \text{if } Z(A, B) > \tau_{\text{upper}} \\ \text{NON-DESCENDANT} & \text{if } Z(A, B) < \tau_{\text{lower}} \\ \text{INCONCLUSIVE} & \text{otherwise} \end{cases} \quad (11)$$

**Threshold Selection.** Under the Gaussian approximation for the null distribution,  $\tau = 1.645$  yields 95% confidence;  $\tau = 2.576$  yields 99% confidence. For high-stakes applications (legal disputes, regulatory audits), we recommend  $\tau_{\text{upper}} = 3.0$ , yielding theoretical FPR  $< 0.13\%$ .

**Robustness Guarantees.** Table 3 summarizes detection rates under transformations.

**Failure Modes.** The protocol cannot provide a verdict when: (1) architectures differ (returns INCOMPATIBLE); (2) perturbation is extreme enough to destroy utility (e.g., 2-bit quantization); (3) both models descend from a common ancestor not available for comparison; (4) adversarial evasion specifically targets the fingerprint (analyzed in Section 6).

**Comparison with Watermarking.** Unlike watermarking (Uchida et al. 2017; Adi et al. 2018), our protocol requires no insertion during training, enables retroactive verification, and survives fine-tuning. The key distinction: watermarks embed artificial signals removable without affecting function, while diagonal dominance arises from optimization itself and cannot be removed without degrading model utility.

## 5 Experiments

Our experiments are organized around four questions that validate the core provenance claim in increasing order of difficulty: the fingerprint must first exist, then generalize, then survive realistic checkpoint modifications, and finally discriminate true weight-level descendants from non-

descendants. We defer adaptive evasion, reparameterization attacks, and boundary cases such as distillation and shared ancestry to Section 6.

Throughout this section, we evaluate the diagonal-dominance score

$$s(M) = \frac{|\text{tr}(M)|}{\|M\|_F}, \quad (12)$$

where  $M$  is an architecture-aware residual branch product. For model-level verification, we use the centered diagonal fingerprint and lineage score defined in Section 4. Unless otherwise stated, reported results include multiple random seeds and confidence intervals.

### 5.1 Exists: Does Residual Training Induce a Diagonal-Dominance Fingerprint?

**Purpose.** This experiment tests the core scientific claim: residual training induces a measurable diagonal-dominance fingerprint in weight space. The goal is to rule out simpler explanations, including initialization artifacts, compatible matrix shapes, residual architecture alone, or the scoring function itself.

**Experimental setup.** We train depth-24 residual MLPs (hidden  $h=64$ , input/output dimension  $d=24$ ) on a synthetic regression target for 200 epochs from four initialization schemes (orthogonal, Kaiming-normal, Xavier-normal, and small Gaussian  $\sigma=0.02$ ), three random seeds each. For each trained checkpoint we compute the score matrix  $S_{ij} = |\text{tr}(W_{\text{out}}^{(j)} W_{\text{in}}^{(i)})| / \|W_{\text{out}}^{(j)} W_{\text{in}}^{(i)}\|_F$ , score Hungarian matching pair accuracy, AUC of correct- vs incorrect-pair scores, and the fraction of correctly paired branch products with negative trace (the dynamic-isometry signature).

**Null model: untrained networks have no fingerprint.** Under Kaiming initialization with no training (depth 48,  $h=96$ ,  $d=48$ ), Hungarian matching achieves pair accuracy of 6.25% (chance is  $1/48 \approx 2.08\%$ ), pair separation  $-0.42$ , and correct- vs incorrect-pair score means that are statistically indistinguishable. The trained counterpart of the same architecture reaches 100% pair accuracy, separation  $+1.18$ , and a  $21\times$  ratio between correct- and incorrect-pair means. The signal is *learned*, not an artifact of matrix shapes or residual connections.

**Non-residual control: skip connections are necessary.** A PlainNet without skip connections, trained on the same task with the same shapes, fails to develop the fingerprint:  $\text{pair\_acc} = 2.8\% \pm 1.8\%$  (chance  $\approx 4\%$ ) and  $\text{AUC} = 0.510 \pm 0.001$ . The matched residual network achieves  $\text{pair\_acc} = 100\% \pm 0\%$  and  $\text{AUC} = 0.984 \pm 0.005$  under identical training. The skip connection itself is the structural prerequisite for the diagonal-dominance fingerprint to emerge during gradient descent.

**Initialization ablation: training enforces isometry from non-isometric starts.** A skeptical reading of dynamical-isometry theory is that the fingerprint requires the network to begin training near orthogonal initialization. We re-train 24-block residual MLPs from four initialization schemes (3

| Init scheme            | Pair acc        | AUC   | Neg. trace | Eval loss |
|------------------------|-----------------|-------|------------|-----------|
| Orthogonal             | $82\% \pm 20\%$ | 0.947 | 100%       | 0.87      |
| Kaiming-normal         | $97\% \pm 4\%$  | 0.981 | 100%       | 1.35      |
| Xavier-normal          | $100\% \pm 0\%$ | 0.990 | 100%       | 0.96      |
| Gaussian $\sigma=0.02$ | $13\% \pm 10\%$ | 0.671 | 32%        | 0.12      |

Table 4: Initialization ablation on depth-24 residual MLPs (3 seeds, 200 epochs each). Three of the four schemes converge to a clean fingerprint with 100% negative trace—training enforces dynamical isometry from any non-trivial starting point. The small-Gaussian init is a clean negative control: with such small initial weights the residual blocks collapse to near-zero contribution and the network shortcuts the task through the skip path, so no Jacobian condition develops and the fingerprint does not appear.

seeds each, 200 epochs, identical optimizer). Table 4 summarizes the results.

Three of four schemes converge to  $\geq 82\%$  pair accuracy with 100% negative trace in the trained branch products. The  $\sigma=0.02$  small-Gaussian case is a clean failure mode: with such small initial weights, the residual blocks collapse to near-zero contribution and the network shortcuts the regression task through the skip path and final read-out. Without active per-block contribution there is no Jacobian condition to enforce, and the fingerprint never develops. This regime is incompatible with deployed language models: GPT-2, BERT, Mistral, and LLaMA-2 all use  $\sigma=0.02$  initialization (Section 5.2) yet recover 100% pair accuracy because the language-modeling objective is hard enough to force non-degenerate residual contribution.

**Training dynamics.** On Park’s 48-block puzzle network, the fingerprint emerges within the first few epochs of training. At epoch 5 the diagonal is fully separated and Hungarian matching recovers 100% of correct pairs. The mean correct-pair score rises monotonically from 0.19 at epoch 0 to 2.07 at epoch 300, while the mean incorrect-pair score remains flat at  $\approx 0.16$ .

**Verdict.** The fingerprint exists, requires both training and skip connections, and develops across initialization schemes whenever residual blocks are driven into non-degenerate use. Untrained networks, PlainNet, and small-Gaussian-init networks (where the residual blocks collapse) all fail to produce the signal.

### 5.2 Generalizes: Does the Fingerprint Transfer Across Architectures?

**Purpose.** This experiment tests whether the fingerprint is a general property of trained residual branches or a narrow artifact of a single architecture, and whether architecture-aware factorization is necessary for extracting the signal.

**Experimental setup.** We evaluate the fingerprint across language transformers (GPT-2 family at four scales; BERT-base; Mistral-7B; LLaMA-2-7B base and chat; Qwen2.5-7B; DeepSeek-R1-Distill-Llama-8B), vision transformers

(ViT-B/16), modern ConvNets (ConvNeXt-T), and ImageNet ResNets (ResNet-50/101/152, Bottleneck blocks). For each architecture, the branch product  $M$  uses the factorization appropriate to that block:  $M = W_2W_1$  for residual MLP and ViT/BERT MLP;  $M = W_3W_2W_1$  for ResNet Bottleneck;  $M = W_OW_V$  for attention V/O paths;  $M = W_QW_K^T$  for attention Q/K paths. For SwiGLU MLPs in Mistral/LLaMA-2/Qwen/DeepSeek we score three architecture-aware factorizations ( $W_{\text{down}}W_{\text{up}}$ ,  $W_{\text{down}}W_{\text{gate}}$ , and the joint stack), and we expand grouped-query-attention  $W_K, W_V$  projections to the query-head count before scoring.

**Transformer families.** Table 5 reports cross-family results. The fingerprint recovers all layers exactly on 27 of 29 evaluated paths across six transformer families spanning four post-training regimes (pretraining, RLHF chat tuning, reasoning distillation, and quantization-distilled variants).

**Three architectural confounds ruled out at once.** The cross-family sweep rules out (i) GELU-activation specificity, since SwiGLU works across four SwiGLU families; (ii) causal-mask specificity, since BERT’s bidirectional attention and Mistral’s sliding-window attention both work; and (iii) full-attention specificity, since 4:1, 7:1, and 8:1 grouped-query attention all give AUC 1.000 on  $W_OW_V$ . Whisper extends the sweep to the speech modality, adding a fourth confound rejection: encoder–decoder cross-attention. Across Whisper-tiny ( $n=4$ ), -base ( $n=6$ ), and -small ( $n=12$ ), both the bidirectional encoder *and* the causal decoder achieve 100% pair accuracy on every path (MLP,  $W_OW_V$ , and the  $Q-K^T$  control)—the fingerprint is independent of modality and of the encoder–decoder architectural split.

**Post-training regime invariance.** The LLaMA-architecture sub-comparison provides a three-regime ablation on identical residual block structure: pretraining (LLaMA-2-7B), RLHF chat tuning (LLaMA-2-7B-chat), and reasoning distillation from R1 (DeepSeek-R1-Distill-Llama-8B). All three preserve the fingerprint at 100% pair accuracy on every attention path and on the joint SwiGLU stack, with  $W_OW_V$  pair separation varying from +0.141 (base) to +0.036 (chat) to +0.280 (distill)—RLHF perturbs the attention output projection more than reasoning distillation, but neither destroys the fingerprint.

**Graceful degradation through factorization redundancy.** The two models with sub-100% sub-paths are both 2024-era SwiGLU models trained with more aggressive schedules than the 2023 baselines. On both, the up-projection path  $W_{\text{down}}W_{\text{up}}$  becomes unusually strong (separation +2.869 for Qwen, +1.469 for DeepSeek; the 2023 baselines have +0.5), and the joint score combining both factorizations recovers all layers exactly. The fingerprint thus degrades gracefully: when one architecture-aware sub-path weakens, the joint factorization captures the union.

**Scaling with hidden dimension.** Across the GPT-2 family the mean correct-pair score grows monotonically with hidden dimension:  $d=768$  at  $\bar{S}_{ii} = 4.18$ ,  $d=1024$  at 5.46,

$d=1280$  at 6.98,  $d=1600$  at 7.77. The fitted scaling matches the  $\Theta(\sqrt{d})$  prediction from the margin analysis (Proposition 1) within  $\leq 8\%$  across the family.

**ImageNet ResNets require architecture-aware factorization.** For ResNet Bottleneck blocks, the naive endpoint product  $W_3W_1$  achieves only chance-level pair accuracy. The architecture-aware triple product  $W_3W_2W_1$  recovers 91–100% of correct pairs across stages of up to 35 blocks. ResNet-50/layer3 ( $n=5$ ) reaches 100% with AUC 1.000; ResNet-101/layer3 ( $n=22$ ) reaches 100% with AUC 0.995; ResNet-152/layer3 ( $n=35$ ) reaches 91% with AUC 0.964. Random-initialization baselines remain at chance even with the correct factorization, confirming that the signal is training-induced and BatchNorm-compatible.

**Verdict.** The fingerprint generalizes across language, vision, and modern ConvNet residual architectures, across four transformer post-training regimes, across initialization schemes, and across two hidden-dimension orders of magnitude. The dominant failure mode (gate-only sub-path on 2024-era SwiGLU models) is rescued by ensemble factorization. Architecture-aware factorization is essential—the naive endpoint product fails on deep ResNet Bottleneck blocks.

### 5.3 Survives: Does the Fingerprint Persist Under Post-Training Transformations?

**Purpose.** This experiment tests whether the fingerprint remains detectable after common checkpoint modifications. This is necessary for practical provenance verification, because real model descendants are often fine-tuned, quantized, pruned, adapted, or otherwise modified before redistribution.

**Experimental setup.** Starting from a depth-24 residual MLP reference  $A$  (200 epochs,  $\text{eval\_loss}(A) = 0.69$ ), we construct transformed descendants  $B$  via five attack families: (i) same-target fine-tuning (50 epochs on a perturbed dataset); (ii) different-target fine-tuning (50 epochs on a different synthetic regression target); (iii) Gaussian weight noise with  $\sigma_{\text{rel}} \in [1\%, 15\%]$ ; (iv) magnitude pruning at sparsities  $\{10, 30, 50, 70, 85\}\%$ ; (v) fake-int8 quantization at  $\{256, 128, 64, 32, 16\}$  levels per matrix. For each transformed  $B$  we compute Hungarian-aligned block-level pair accuracy on the branch product  $M = W_{\text{out}}W_{\text{in}}$ , model-level lineage score  $\mathcal{L}(A, B)$  (Eq. 9), and downstream utility (eval MSE).

**Block-level pairing survival.** Across all 75 descendants (3 references  $\times$  25 transformations), Hungarian matching recovers 100% of the original block correspondences in 74/75 cases; the single prune(85%) run drops to 96% when the sparsity destroys most of the smallest-magnitude entries. Block-level pairing is essentially preserved under realistic post-training transformations.

**Model-level lineage survival.** Table 6 reports the model-level lineage score  $\mathcal{L}(A, B)$  by transformation type. The score remains far above any independently-trained baseline ( $\mathcal{L} \leq 0.084$  for  $n=45$  same-task independents and  $\leq 0.086$



| Model                        | Family / training                 | $L$                   | Path           | Pair acc    | AUC       |
|------------------------------|-----------------------------------|-----------------------|----------------|-------------|-----------|
| GPT-2 (124M–1.5B)            | causal LM, GELU                   | 12–48                 | MLP+V/O+Q/K    | <b>100%</b> | 1.000     |
| BERT-base                    | encoder, MLM, GELU                | 12                    | MLP+V/O+Q/K    | <b>100%</b> | 0.97–1.00 |
| ViT-B/16                     | vision encoder                    | 12                    | MLP+V/O+Q/K    | <b>100%</b> | 1.000     |
| ConvNeXt-T stage 3           | modern ConvNet                    | 9                     | MLP            | <b>100%</b> | 0.875     |
| Mistral-7B                   | SwiGLU+GQA(4:1)+SWA, 2023         | 32                    | MLP×3+V/O+Q/K  | <b>100%</b> | 0.99–1.00 |
| LLaMA-2-7B                   | SwiGLU+full MHA, pretrain         | 32                    | MLP×3+V/O+Q/K  | <b>100%</b> | 0.97–1.00 |
| LLaMA-2-7B-chat              | +RLHF instruction tuning          | 32                    | MLP×3+V/O+Q/K  | <b>100%</b> | 0.96–1.00 |
| Qwen2.5-7B                   | SwiGLU+GQA(7:1)+biases, 2024      | 28                    | MLP×3+V/O+Q/K  | 4/5 + joint | 0.94–1.00 |
| DeepSeek-R1-Distill-Llama-8B | +R1 reasoning distillation, 2024  | 32                    | MLP×3+V/O+Q/K  | 4/5 + joint | 0.93–1.00 |
| Whisper-tiny/base/small      | speech enc–dec, GELU              | 4+4 to 12+12          | MLP+V/O+Q/K    | <b>100%</b> | 0.85–1.00 |
| ResNet-50/101/152            | BatchNorm, Bottleneck $W_3W_2W_1$ | up to 35 blocks/stage | triple product | 91–100%     | 0.96–1.00 |

Table 5: Cross-architecture fingerprint generalization. Pair accuracy is fraction of layers correctly recovered by Hungarian matching on the architecture-aware branch product. The two sub-100% paths (Qwen2.5 and DeepSeek  $W_{\text{down}}W_{\text{gate}}$ , at 68% and 84% respectively) are both rescued to 100% by the joint SwiGLU score  $W_{\text{down}}[W_{\text{up}}; W_{\text{gate}}]$ . All random-initialization baselines collapse to  $\leq 9\%$  pair accuracy.

| Transformation                                   | $n$ | Mean $\mathcal{L}$ | Min $\mathcal{L}$ | Det. @ 1%FPR |
|--------------------------------------------------|-----|--------------------|-------------------|--------------|
| Fine-tune (same target)                          | 15  | 0.996              | 0.990             | 15/15        |
| Quantization (16–256 lvl)                        | 15  | 0.995              | 0.980             | 15/15        |
| Gauss. noise ( $\sigma_{\text{rel}} \leq 15\%$ ) | 15  | 0.993              | 0.974             | 15/15        |
| Fine-tune (diff target)                          | 15  | 0.944              | 0.840             | 15/15        |
| Mag. pruning (10–85%)                            | 15  | 0.810              | 0.575             | 15/15        |
| <i>Indep. (same task)</i>                        | 45  | 0.084              | 0.022             | —            |

Table 6: Survival of the model-level lineage score under post-training transformations. Even the weakest descendant transformation (heavy pruning) keeps  $\mathcal{L} \geq 0.58$ , an order of magnitude above the independent baseline mean of 0.084. All 75 descendants are detected at the 1%FPR operating point.

for  $n=9$  distilled students) under every transformation we test.

**Utility-fingerprint coupling.** The transformations are ordered by impact on both lineage and utility. Heavy pruning (85% sparsity) simultaneously drives lineage to  $\mathcal{L}=0.58$  and eval loss to 1.4 ( $2\times$  degradation); aggressive quantization (16 levels) keeps lineage at  $\mathcal{L}=0.98$  and eval loss within 5% of the reference; mild fine-tuning (50 epochs, same target) leaves both at the reference level. Across the 75 descendants, lineage score is rank-correlated with eval loss ( $\rho = -0.83$  at the per-descendant level, computed across all attack types): when the fingerprint weakens, the model itself has been damaged.

**RLHF fine-tuning preserves the fingerprint on production transformers.** On real production checkpoints, we evaluate the LLaMA-2-7B base  $\rightarrow$  LLaMA-2-7B-chat transition, which corresponds to substantial RLHF and instruction-tuning post-training. All five branch-product paths (MLP  $W_{\text{down}}W_{\text{up}}$ ,  $W_{\text{down}}W_{\text{gate}}$ , joint, and the two attention products) survive at 100% pair accuracy. Pair separation matches the base model to within  $\pm 0.005$  on four of five paths; only  $W_O W_V$  shows a measurable drop ( $+0.141 \rightarrow$

$+0.036$ ), indicating that RLHF perturbs the attention output projection more than the MLP or Q/K paths but does not destroy the diagonal mass.

**Verdict.** The fingerprint survives the full battery of common post-training transformations—fine-tuning across both same-target and target-shifted objectives, quantization down to 16 levels, magnitude pruning up to 85% sparsity, weight noise up to  $\sigma_{\text{rel}}=15\%$ —with  $\mathcal{L} \geq 0.58$  in every case and detection at the 1% FPR operating point in 75/75 cases. RLHF fine-tuning on LLaMA-2 preserves the fingerprint at 100% pair accuracy on every path. Fingerprint degradation, when it occurs, coincides with measurable utility loss.

#### 5.4 Discriminates: Can the Fingerprint Distinguish Descendants from Non-Descendants?

**Purpose.** This experiment tests the central provenance claim: the fingerprint should distinguish true weight-level descendants from unrelated or merely functionally similar models.

**Experiment.** We construct a controlled benchmark of reference–suspect checkpoint pairs on depth-24 residual MLPs trained on synthetic regression targets. *Positive* pairs share weight-level lineage and span five attack types: fine-tuning on the same target ( $n=15$ ); fine-tuning on a different target ( $n=15$ ); Gaussian weight noise with relative magnitudes 1%–15% ( $n=15$ ); magnitude pruning with sparsities 10%–85% ( $n=15$ ); and fake-int8 quantization at 16–256 levels ( $n=15$ ). *Negative* pairs do not share weight-level lineage and span four families: 15 independently trained models on the same task ( $A \times 15 = 45$  pairs); 5 models trained on different tasks; 5 randomly initialized models; and 3 distilled students trained to mimic each reference’s outputs (the most stringent negative control: function-similar but weight-independent).

For each pair  $(A, B)$ , we compute the aligned residual signatures (Eq. 7), the model-level lineage score  $\mathcal{L}(A, B)$  (Eq. 9), and per-reference z-scores  $Z(A, B)$ . We calibrate

thresholds using the empirical null distribution of unrelated checkpoints.

**Results.** Table 2 reports score distributions by suspect category. The proposed residual-signature score achieves **AUROC** = 1.000, **AUPRC** = 0.987, **TPR@1%FPR** = 100%, **TPR@10%FPR** = 100% across 84 descendant-non-descendant test pairs per reference (252 total). Every descendant in every attack category is detected at the 1%FPR operating point. Conversely, 0/45 same-task independents and 0/15 different-task independents are falsely classified as descendants. The hardest negative control—distilled students that learn to mimic the reference’s outputs but train fresh weights—produces a single false positive (1/9 at 1%FPR), confirming the score tracks weight-level lineage rather than function-level similarity.

**Branching ancestry recovery.** We additionally test the score’s ability to recover multi-generation ancestry on a synthetic branching tree  $C_1 \rightarrow \{C_2, C_3\}$ ,  $C_2 \rightarrow C_4$ ,  $C_3 \rightarrow C_5$  plus 5 independent same-architecture controls  $\{I_0, \dots, I_4\}$  (training details in supplementary). For each leaf in  $\{C_4, C_5\}$ , the lineage score ranks candidates in the correct ancestry order: parent > grandparent > uncle > cousin > independent. Concretely, for  $C_4$  (true parent  $C_2$ , true grandparent  $C_1$ ):

$$\begin{aligned} \mathcal{L}(C_4, C_2) &= 0.961 > \mathcal{L}(C_4, C_1) = 0.910 \\ &> \mathcal{L}(C_4, C_3) = 0.867 > \mathcal{L}(C_4, C_5) = 0.850 \\ &\gg \mathcal{L}(C_4, I_k) \approx 0.08. \end{aligned}$$

The score decays monotonically with genealogical distance: direct parent (0.95), grandparent (0.91), sibling (0.90), uncle (0.87), cousin (0.85), independent (0.08). Critically,  $\mathcal{L}(C_4, C_5) = 0.850 \gg \mathcal{L}(C_4, \text{independent}) = 0.081$ : cousins that never saw each other’s training data—they diverged at  $C_1$  through entirely different fine-tune branches—still inherit a detectable shared residual signature from their common ancestor. This demonstrates that the score supports compositional reasoning about checkpoint ancestry rather than only binary descendant detection.

**Baseline ablation.** To verify that the residual-signature subtraction  $R_\ell = M_\ell - \alpha_\ell^* I$  is doing real work, we compare against the scalar diagonal-dominance score  $s(M_\ell) = |\text{tr}(M_\ell)| / \|M_\ell\|_F$  applied directly as a lineage statistic,  $\text{DiagOnly}(B) = \frac{1}{L} \sum_\ell s(M_\ell^B)$ . This baseline achieves AUROC = 0.417 (chance level) on the same benchmark: it can detect whether  $B$  is a trained residual model but cannot distinguish a descendant of  $A$  from an independently trained same-architecture model. Removing the generic identity component before comparison is essential.

**Real architecture on natural images.** To verify that discrimination is not an artifact of the synthetic depth-24 MLP benchmark, we replicate the experiment on a real convolutional architecture and natural images. We train two CIFAR-10 ResNet-18 references and three same-architecture independents from different seeds. The residual signature is extracted per BasicBlock from the BN-folded branch product  $M_\ell = \text{vec-sum}(W_\ell^{(2)}) \text{vec-sum}(W_\ell^{(1)})$ , where vec-sum

projects the  $3 \times 3$  convolution to its  $(C_{\text{out}}, C_{\text{in}})$  channel matrix by summing over kernel positions; block dimensions are heterogeneous across stages ( $\{64, 64, 128, 256, 512\}$ ), so the Hungarian alignment is replaced by canonical block-index matching. Each reference yields 11 cheap descendants (Gaussian weight noise  $\sigma_{\text{rel}} \in \{1-10\%\}$ , magnitude pruning at 20–80%, and fake-int8 quantization at 16–256 levels). The score achieves **AUROC** = 1.000 and **TPR@1%FPR** = 100% over the 22 descendants vs. 6 independent pairs. Descendant scores span  $\mathcal{L} \in [0.695, 1.000]$  across attack types (noise mean 0.983, quantization mean 0.954, pruning mean 0.864), while the same-architecture independent baseline tops out at  $\mathcal{L}_{\text{max}} = 0.004$  (mean 0.0014). The separation gap exceeds two orders of magnitude even between the worst-case descendant (80% pruning) and the best-case independent, confirming that the residual signature transfers to convolutional architectures and that the discrimination is driven by weight-level provenance rather than architecture-induced isometry structure.

**Exit Criteria.** This question is answered positively. True descendants score consistently above the calibrated threshold ( $\mathcal{L} \geq 0.81$  even under aggressive transformations); non-descendants remain near the null distribution ( $\mathcal{L} \leq 0.20$ ); independently trained same-architecture models and distilled students do not falsely appear as descendants; and the score recovers multi-generation ancestry orderings on branching trees.

## 6 Adaptive Evasion and Boundary Cases

A residual-signature lineage score is only useful if it cannot be cheaply suppressed by an adversary who wants to disguise a stolen model. We test three classes of attack: function-preserving reparameterizations (which form a basis-dependence objection), weight-space rotations (a relaxed attack that targets the weight product directly), and gradient-based direct suppression. Across all three, the score is either exactly invariant or can only be suppressed by destroying the model.

### 6.1 Function-Preserving Reparameterizations

**Theory.** A natural function-preserving symmetry of a ReLU residual block is per-block permutation of hidden units: for permutation matrix  $P_\ell$  of size  $h \times h$ , the substitution  $W_{\text{in},\ell} \leftarrow P_\ell W_{\text{in},\ell}$ ,  $W_{\text{out},\ell} \leftarrow W_{\text{out},\ell} P_\ell^\top$  leaves the block’s output exactly unchanged because  $P_\ell$  commutes with the elementwise ReLU. The branch product satisfies

$$W_{\text{out},\ell} P_\ell^\top \cdot P_\ell W_{\text{in},\ell} = W_{\text{out},\ell} \cdot W_{\text{in},\ell},$$

so  $M_\ell$ , the residual signature  $\phi_\ell$ , and consequently the lineage score  $\mathcal{L}$  are exactly invariant.

**Empirical confirmation.** We apply independent random permutations to each of the 24 blocks of a trained reference checkpoint  $A$ , repeated for 5 seeds. As predicted, every run produces  $\mathcal{L}(A, P(A)) = 1.0$  to numerical precision and the model output is identical to  $A$  to within floating-point error ( $\|f(A) - f(P(A))\|_2^2 < 2 \times 10^{-13}$ ). The function-preserving permutation symmetry cannot evade detection.

| $\lambda$ | Final $\mathcal{L}$ | Util drift $\ \Delta f\ ^2$ | Eval loss | Verdict            |
|-----------|---------------------|-----------------------------|-----------|--------------------|
| 0         | 0.053               | 38.7                        | 39.5      | utility destroyed  |
| $10^{-2}$ | 0.054               | $7.4 \times 10^{-2}$        | 0.77      | utility +12% loss  |
| $10^{-1}$ | 0.083               | $3.2 \times 10^{-2}$        | 0.70      | utility +1.5% loss |
| $10^0$    | 0.908               | $2.2 \times 10^{-4}$        | 0.69      | utility preserved  |
| $10^1$    | 0.963               | $1.7 \times 10^{-7}$        | 0.69      | utility preserved  |
| $10^2$    | 0.963               | $5.1 \times 10^{-8}$        | 0.69      | utility preserved  |

Table 7: Gradient-attack Pareto frontier on a depth-24 residual MLP reference. As the utility-preservation weight  $\lambda$  increases, the attacker is forced to leave the fingerprint intact ( $\mathcal{L} \rightarrow 0.96$ ); to drive the score down to the independent-baseline level  $\mathcal{L} \approx 0.08$ , the attacker must produce a utility drift of  $7 \times 10^{-2}$  or worse—increasing eval loss by  $\sim 12\%$  from a reference loss of 0.69. The reference’s self-similarity is  $\mathcal{L}(A, A) = 1.000$  and an independently-trained same-architecture baseline has  $\mathcal{L} = 0.084$ .

## 6.2 Orthogonal-Rotation Attacks

**Theory.** Generalizing permutations to arbitrary orthogonal matrices  $R_\ell \in O(h)$  preserves the weight product— $W_{\text{out},\ell} R_\ell^\top R_\ell W_{\text{in},\ell} = W_{\text{out},\ell} W_{\text{in},\ell}$ —but *not* the block’s output, because  $R_\ell$  does not in general commute with ReLU. An adversary willing to accept arbitrary function change can apply per-block rotations.

**Empirical confirmation.** We sample 5 independent rotation sets and apply them to  $A$ . The lineage score  $\mathcal{L}(A, R(A)) = 1.0$  to numerical precision in every run, exactly as predicted—the rotated model has the same fingerprint as  $A$  even though its outputs differ by  $\|f(A) - f(R(A))\|^2 \approx 2$  (a meaningful functional change). The residual-signature score is thus a property of the weight product itself, not of the block’s function.

## 6.3 Direct Fingerprint-Suppression Attacks

The remaining adaptive attack: an adversary directly optimizes a weight perturbation  $\Delta$  to minimize the lineage score subject to a utility constraint. We solve

$$\min_{\Delta} \mathcal{L}_{\text{cos}}(A, A + \Delta) + \lambda \cdot \|f_{A+\Delta}(X_{\text{eval}}) - f_A(X_{\text{eval}})\|_2^2, \quad (13)$$

where  $\mathcal{L}_{\text{cos}}$  is the differentiable mean-cosine relaxation of the lineage score (skipping the non-differentiable Hungarian alignment, which is unnecessary when block correspondence is preserved). Sweeping  $\lambda \in \{0, 10^{-2}, 10^{-1}, 1, 10, 10^2\}$  traces the Pareto frontier between utility-preservation and lineage-suppression.

The Pareto curve (Table 7) decomposes cleanly. With  $\lambda \geq 1$ , the attacker preserves utility (drift  $\leq 2 \times 10^{-4}$ , eval loss matches reference) but cannot push  $\mathcal{L}$  below 0.91—an order of magnitude above any non-descendant in our benchmark. To suppress  $\mathcal{L}$  to chance level, the attacker must accept utility damage of at least  $\|\Delta f\|^2 \approx 7 \times 10^{-2}$ , which translates to a 12% eval-loss penalty. At  $\lambda = 0$  (utility unconstrained), the attacker succeeds at fingerprint suppression but destroys the model entirely (eval loss 39.5, a  $57\times$  degradation).

**Implication.** The diagonal-dominance fingerprint inherits its robustness from the same dynamic-isometry condition that makes the model useful: any perturbation large enough to disrupt  $W_{\text{out},\ell} W_{\text{in},\ell} \approx -\varepsilon I + E_\ell$  also damages gradient flow through the residual stream. There is no free utility-preserving region in which an adversary can suppress the fingerprint.

## 6.4 Distillation and Retraining

The fingerprint tracks weight-level lineage, not function-level imitation. This is a feature, not a bug: it distinguishes our method from behavioral fingerprints that confuse a knowledge-distilled student with the teacher.

**Empirical confirmation.** In the §5.4 benchmark, we trained 9 distilled students that mimic the soft outputs of each reference checkpoint. Each student had fresh randomly-initialized weights and learned only from the teacher’s outputs (no shared weights). Their mean lineage score is  $\mathcal{L} = 0.086$ , statistically indistinguishable from same-architecture independently-trained models ( $\mathcal{L} = 0.084$ ). Only 1/9 distilled students leaks above the 1%-FPR threshold—the same false-positive rate as independent training. Distillation, retraining, and independent training are all correctly classified as non-descendants.

This boundary is the right one for provenance verification: if an adversary distills  $A$  into a student with fresh weights, they have not stolen  $A$ ’s weights and our method correctly says so. Conversely, weight-level descendants—fine-tuned, pruned, quantized, LoRA-merged—are reliably detected because they inherit  $A$ ’s residual structure.

## 6.5 Function-Preserving Cross-Layer Rescaling

A subtler attack relies on cross-layer rescaling: for any positive scalar  $c$ , the substitution  $W_{\text{in},\ell} \leftarrow c \cdot W_{\text{in},\ell}$ ,  $W_{\text{out},\ell} \leftarrow W_{\text{out},\ell}/c$  is a function-preserving symmetry of the unbiased ReLU block (ReLU is positive-homogeneous). The branch product satisfies  $(W_{\text{out}}/c)(cW_{\text{in}}) = W_{\text{out}}W_{\text{in}}$ , so  $M_\ell$  is unchanged and  $\mathcal{L}$  is exactly invariant. With non-zero  $b_{\text{in}}$ , the symmetry also requires  $b_{\text{in}} \leftarrow c \cdot b_{\text{in}}$ . Our residual-signature score does not look at biases, so the attack leaves  $\mathcal{L} = 1$  regardless. The remaining basis-dependence objections—invertible non-orthogonal basis changes that mix layers—are not function-preserving for a ReLU block and require optimization (Section 6.3).

## 7 Related Work

**Purpose:** Situate the contribution without bloating the paper.

**Recommended subsections:**

### 1. Neural Network Watermarking

- Embedded parameter watermarks
- Backdoor/trigger watermarks
- Watermark capacity
- NeuralMark / robust weight-based watermarking

### 2. Model Fingerprinting and Ownership Verification

- Black-box fingerprints
- Decision-boundary fingerprints
- Intrinsic/statistical fingerprints

### 3. Training Dynamics and Residual Networks

- Dynamical isometry
- Residual near-identity behavior
- Weight-space structure

### 4. Model Provenance and Trustworthy ML

- Model supply chains
- Checkpoint auditability
- Lineage verification

## 8 Discussion and Limitations

**Scope.** Our method is a white-box auditing tool: it requires access to the suspect checkpoint’s weights, and it presumes residual architecture. The first restriction is intrinsic—there is no recoverable weight-level signature without weights. The second tracks where the diagonal-dominance phenomenon comes from: residual connections force the branch product toward  $-\varepsilon I + E_\ell$  for stable gradient flow, and non-residual networks (e.g., classic feed-forward stacks) have no analogous identity-target constraint. The fingerprint should therefore not be expected on non-residual designs.

**Weight-level, not behavioral.** The diagonal-dominance signature is a property of the trained weight matrices themselves, not of the function the model computes. Two consequences follow. First, the method correctly classifies as non-descendant any model that was trained from scratch—even one trained to mimic a reference via distillation ( $\mathcal{L} \approx 0.086$ , indistinguishable from the same-architecture independent baseline at 0.084; §6.4). Second, the method cannot adjudicate behavioral imitation, knowledge transfer, or training-data overlap. If the question is “did this checkpoint inherit this reference’s weights?” our method answers it; if the question is “does this checkpoint encode the same knowledge as the reference?” it does not, and behavioral fingerprints or query-based attribution remain appropriate.

**Adaptive reparameterization requires care.** Two function-preserving symmetries leave  $\mathcal{L} = 1$  exactly: per-block hidden-unit permutations and positive cross-layer rescaling (§6.1, §6.5). Per-block orthogonal rotations preserve the branch product but not the function, and also yield  $\mathcal{L} = 1$  (§6.2). These are invariances of our score, not vulnerabilities: they imply the method is insensitive to the underlying basis. The class of attacks that genuinely threatens the score—direct gradient-based suppression—falls cleanly on the wrong side of a Pareto frontier (Table 7): driving  $\mathcal{L}$  to the independent-baseline level requires at least a 12% utility loss, and full chance-level suppression destroys the model entirely ( $57\times$  eval-loss degradation). We have not characterized all possible adaptive attacks, however, and a determined adversary willing to retrain or distill can always escape detection by producing a model that is, by our definition, not a weight-level descendant.

**Positioning.** Our method is not a proof of safety: a checkpoint that passes lineage verification is not certified clean, only weight-attributable to a reference. It is also not a replacement for cryptographic watermarking. Watermarks remain the right tool when an owner can intervene before deployment and accepts the small utility cost of embedding; diagonal-dominance fingerprints are complementary, offer-

ing passive, retroactive verification for the much larger population of checkpoints that were never watermarked. We see provenance verification as one component of a layered audit stack alongside metadata, behavioral fingerprints, and watermarking—each addressing a different failure mode of the open-weight ecosystem.

## 9 Conclusion

We have shown that training itself can provide provenance signal. Residual-network optimization leaves an intrinsic weight-space trace—a diagonal-dominance fingerprint in the architecture-aware branch product, taking the negative-identity-like form  $W_{\text{out}}W_{\text{in}} \approx -\varepsilon I + E_\ell$  in residual MLPs and a broader trace/Frobenius-concentration form in multi-factor and attention paths. The margin formula derived for the residual-MLP regime predicts a  $\Theta(\sqrt{d})$  separation margin and is followed empirically by GPT-2 MLPs. Across 12 architectures spanning five families, the fingerprint is detectable in trained but not random-initialized models, recovers block correspondences with 91–100% accuracy under five classes of post-training modification, and yields model-level lineage detection at AUROC = 1.000 (TPR = 100% at 1% FPR) on both synthetic depth-24 MLP benchmarks and real ResNet-18/CIFAR-10 checkpoints. In our direct gradient-based suppression experiment, adaptive attacks are forced through a strict utility–detection Pareto frontier on which the attacker cannot win without paying a measurable accuracy cost. Together, these results establish a passive, retroactive route to white-box weight-level lineage verification under common post-training transformations, for suspect checkpoints of compatible residual architecture—one that requires no watermark, no metadata, and no cooperation from the original trainer.

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